

1. Derive the expression for 3-phase apparent power in terms of symmetrical components.

$$S = P + jQ = V_a I_a^* + V_b I_b^* + V_c I_c^* \quad S = \begin{bmatrix} v_a & v_b & v_c \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}^* = \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}^T \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}^*$$

$$S = [A V]^T [AI]^*$$

$$S = V^T A^T A^* I^*$$

Noting that $A^T = A$ and a and a^2 are conjugates,

$$S = \begin{bmatrix} v_{a0} & v_{a1} & v_{a2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix}^*$$

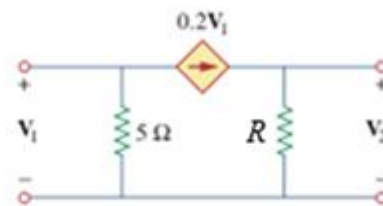
or, since $A^T A^*$ is equal to $3U$ where U is 3×3 unit matrix

$$S = 3 \begin{bmatrix} v_{a0} & v_{a1} & v_{a2} \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix}^*$$

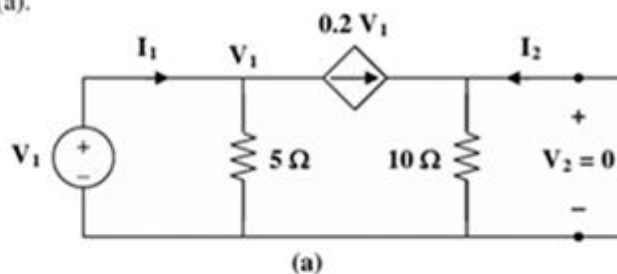
$$S = V_a I_a^* + V_b I_b^* + V_c I_c^* = 3 V_{a0} I_{a0}^* + 3 V_{a1} I_{a1}^* + 3 V_{a2} I_{a2}^*$$

(This problem was given to you for a self-test one day before the quiz.)

2. Find the y parameters of the network shown in Fig. 1. $R = 2, 3, 4, 5, 6, 7 \Omega$.



To get y_{11} and y_{21} , refer to Fig. (a).

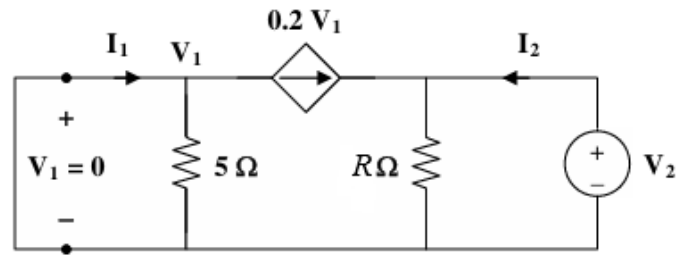


At node 1,

$$I_1 = \frac{V_1}{5} + 0.2V_1 = 0.4V_1 \longrightarrow y_{11} = \frac{I_1}{V_1} = 0.4$$

$$I_2 = -0.2V_1 \longrightarrow y_{21} = \frac{I_2}{V_1} = -0.2$$

To get y_{22} and y_{12} , refer to the circuit in Fig. (b).



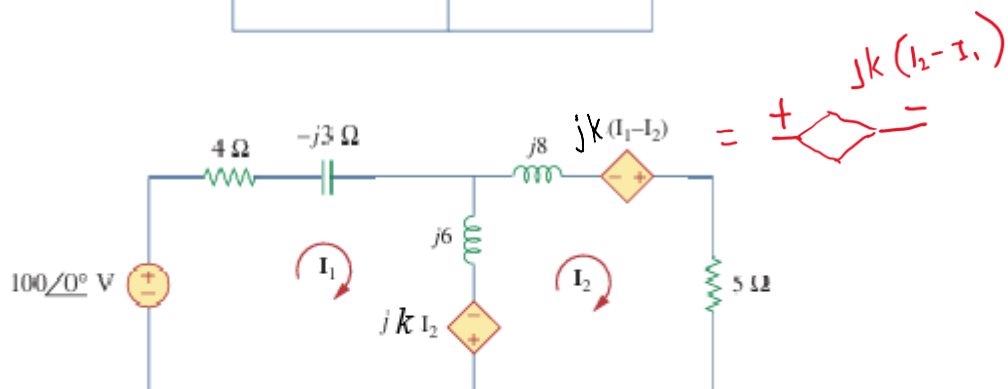
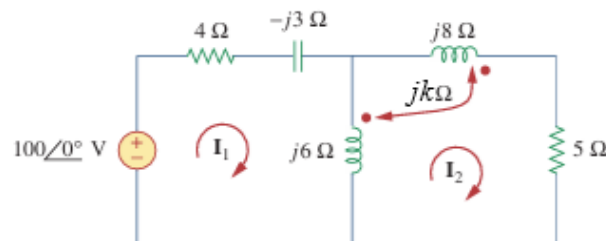
Since $V_1 = 0$, the dependent current source can be replaced with an open circuit.

$$V_2 = RI_2 \longrightarrow y_{22} = \frac{I_2}{V_2} = \frac{1}{R}$$

$$y_{12} = \frac{I_1}{V_2} = 0$$

$$[y] = \begin{bmatrix} 0.4 & 0 \\ -0.2 & \frac{1}{R} \end{bmatrix} S$$

3. Write the mesh equations for the circuit shown in Fig.2. $k = 0.22, 0.12, 0.18, 0.5, 1.2, 0.8$



Mesh equations are

$$(4+j3)I_1 - j(6+k)I_2 = 100$$

$$-j(6+k)I_2 + 5 + j(14+2k)I_2 = 0$$